Nithya Madalakote Prakash

AI Engineer | RAG, LLM Agents and MLOps | Germany

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SUMMARY

M.Sc. Artificial Intelligence student (BTU Cottbus-Senftenberg) who builds RAG, LLM-agent, MLOps and full-stack systems and evaluates them with labelled test sets, baselines and load tests. Prior Java Spring Boot full-stack and data science internship experience. Seeking a Werkstudent or internship role in AI, machine learning or software engineering in Germany.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript, JavaScript

Generative AI: LLMs, RAG, hybrid retrieval (dense + BM25 + RRF), cross-encoder reranking, LangGraph, AI agents, MCP, prompt engineering, LoRA fine-tuning, LLM evaluation (Recall@k, MRR, RAGAS)

Machine Learning: PyTorch, scikit-learn, XGBoost, SHAP, time-series forecasting, anomaly detection, computer vision

Backend and Data: FastAPI, Spring Boot, REST APIs, PostgreSQL, Qdrant, Neo4j, Redis, Celery, Kafka, MQTT

DevOps and Frontend: Docker, Kubernetes, GitHub Actions CI/CD, Prometheus, Grafana, Linux, Git, pytest, React, Next.js, Tailwind CSS

PROJECTS

Industrial AI Copilot | LangGraph, FastAPI, Qdrant, PostgreSQL, Celery, Next.js github.com/nithya-prakash/IndustrialAICopilot

- Built a RAG and LLM maintenance copilot: LangGraph multi-agent supervisor with human approval, hybrid search (BM25 + dense + cross-encoder rerank) over equipment manuals, and z-score sensor fault checks on the real UCI hydraulic rig dataset.
- Evaluated on 53 labelled questions: retrieval MRR 0.959 (hybrid + rerank), citation precision 0.91, 7 of 8 unanswerable questions declined; documented weak spots (valve fault AUROC 0.84).

ClauseGuard | RAG, LLM, LoRA, Qdrant, Neo4j, FastAPI, Next.js github.com/nithya-prakash/ClauseGuard

- Built first-pass compliance screening of German contracts against BGB and DSGVO: hybrid retrieval (RRF + cross-encoder), evidence-sufficiency gate, citation verification, MCP server, Celery worker and local Ollama LLM provider.
- Recall@5 1.00 on 15 retrieval queries; evidence gate 0.79 accuracy on 63 clauses; 135 tests passing. Reported that the LoRA fine-tuned DistilBERT classifier did not beat a TF-IDF baseline (0.56 vs 0.63).

HybridRAG | Python, FastAPI, Qdrant, PostgreSQL, Celery, Next.js, Docker github.com/nithya-prakash/HybridRAG

- Built a multi-user RAG assistant with dense + BM25 retrieval fused by RRF, cross-encoder reranking, citation-aware answers and a guard that declines before generating; deployed a live demo.
- On a 30-question hard set, reranking raised Recall@1 from 0.483 (dense only) to 0.817; hallucination guard 93.1% accuracy; 284 automated tests.

FleetOps | MLOps, MQTT, FastAPI, Prometheus, Docker, Kubernetes github.com/nithya-prakash/FleetOps

- Built MLOps for edge AI: model registry, quality-gated canary rollout with automatic rollback, drift detection and monitoring, trained on the real UCI hydraulic rig data (F1 0.85 on 662 held-out cycles).
- 367 tests plus 52 live security checks and 12 failure-injection scenarios; rollout to 100 simulated devices in 12 s; 50-minute soak test with no memory growth.

InduSense | Python, FastAPI, Kafka, MQTT, Kubernetes, Next.js github.com/nithya-prakash/InduSense

- Built an event-driven industrial IoT platform (MQTT, Kafka, five FastAPI services, Isolation Forest written from scratch, Next.js dashboard) verified on Kubernetes with 148 tests; dashboard p95 65 ms at 50 virtual users.

EXPERIENCE

Data Science Intern | LaunchED Global, Remote (Bengaluru, India) Mar 2025 - Apr 2025

- Worked on data analysis, preprocessing and insight extraction with Python and NumPy in a two-month internship.

Java Full Stack Developer Trainee | Palle Technologies, Bengaluru, India Jul 2024 - Jan 2025

- Built and tested REST APIs with Java, Spring Boot and MySQL and integrated them with responsive front-end components for internal web applications.
- Delivered 8+ production features, wrote 150+ unit tests (87% coverage), resolved 25+ defects and optimized 12+ SQL queries (40% lower API latency); took part in code reviews and Agile ceremonies.

EDUCATION

M.Sc. Artificial Intelligence | Brandenburg University of Technology Cottbus-Senftenberg, Germany Oct 2025 - Present
Coursework: Neural Signal Analysis, Image Processing and Computer Vision, Causal Data Science, Modeling of Perception and Action

B.Tech. Computer Science | Presidency University, Bengaluru, India 2020 - 2024
GPA 7.62 / 10

PUBLICATION

MediSync: Streamlining Healthcare Continuity, IJRPR, ISSN 2582-7421 2023

LANGUAGES AND CERTIFICATES

Languages: English (C1), German (A2, in progress) | **Certificates:** Introduction to Machine Learning (Kaggle); The Complete Data Science Bootcamp (Udemy)